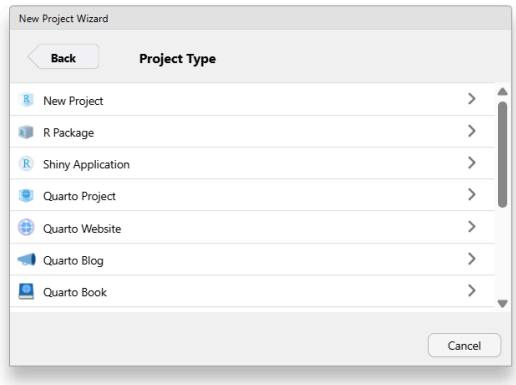
The dataset I used for this project is the **UCBAdmissions** from the base R datasets package. It includes admission results from UC Berkeley, broken down by whether applicants were admitted or rejected, along with their gender and the department they applied to.

I chose the UCBAdmissions dataset because it allows for clear comparisons across categories such as department, gender, and admission status. I also wanted a dataset that would work well with a dropdown menu and a bar chart. This made it a good fit for building an interactive Shiny app while exploring patterns in decision outcomes.

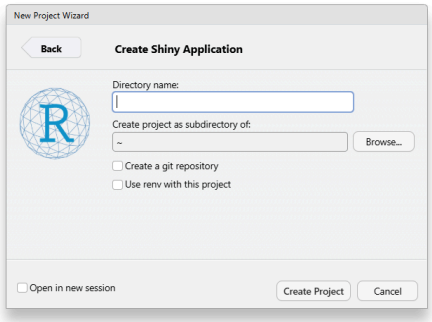
Documentation
I documented the development of the Shiny app by capturing screenshots at key stages of the process.

Step 1: Project Setup

I created a new Shiny application project in RStudio, which automatically generated the app.R file. **File → New Project → New Directory → Shiny Application**



Then I named it ucb_admissions_app.



Step 2: Initial Template

RStudio generated a default Shiny template, which in the next step I replaced with my own code.

```
library(shiny)

# Define UI for miles per gallon app ----
ui <- pageWithSidebar(

  # App title ----
  headerPanel("Miles Per Gallon"),

  # Sidebar panel for inputs ----
  sidebarPanel(),

  # Main panel for displaying outputs ----
  mainPanel()
)

# Define server logic to plot various variables against mpg ----
server <- function(input, output) {

}

shinyApp(ui, server)
```

Step 3: User Interface Creation

I defined the user interface using the tutorial structure, including **pageWithSidebar()**, **headerPanel()**, **sidebarPanel()**, and **mainPanel()**. I also added a title using **headerPanel()** to label the application as UCB Admissions Explorer. Within the sidebar panel, I added a dropdown menu using **selectInput()** so the user can choose a department. In the main panel, I included both **textOutput()** for displaying a dynamic title and **plotOutput()** to display the graph.

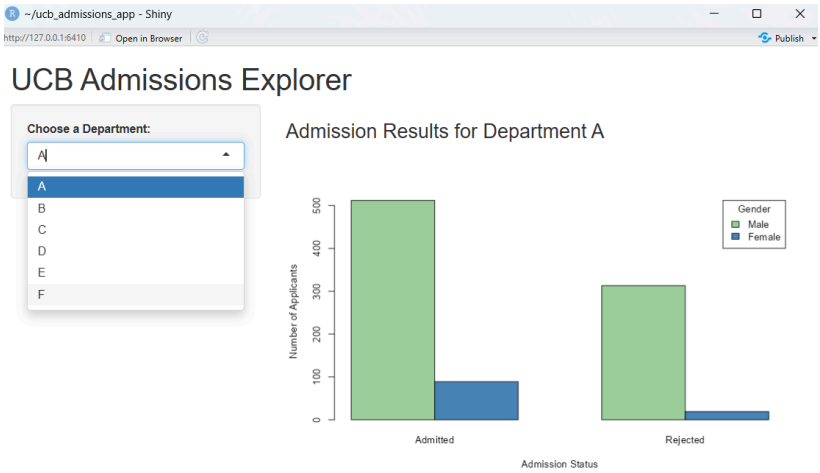
```r
# Define UI for UCBAdmissions app ----
ui <- pageWithSidebar(

  # App title ----
  headerPanel("UCB Admissions Explorer"),

  # Sidebar panel for inputs ----
  sidebarPanel(
    selectInput("department",
                "Choose a Department:",
                choices = c("A", "B", "C", "D", "E", "F"))
  ),

  # Main panel for displaying outputs ----
  mainPanel(
    h3(textOutput("title")),
    plotOutput("admissionsPlot")
  )
)
```

Step 4: Server Function

I created the server function to control how the app responds to user input. I used **renderPlot()** to generate a bar chart that updates based on the selected department. The dataset is filtered using **input$department**, which allows the visualization to change dynamically.

I also added a dynamic title using **renderText()**, which updates to display the selected department at the top of the chart. This makes it clearer to the user which department is being viewed.

```r
# Define server logic ----
server <- function(input, output) {

  output$title <- renderText({
    paste("Admission Results for Department", input$department)
  })

  output$admissionsPlot <- renderPlot({

    dept_data <- subset(admissions, Dept == input$department)

    counts <- xtabs(Freq ~ Gender + Admit, data = dept_data)

    barplot(counts,
            beside = TRUE,
            col = c("darkseagreen3", "steelblue"),
            legend.text = TRUE,
            args.legend = list(x = "topright", title = "Gender"),
            xlab = "Admission Status",
            ylab = "Number of Applicants")
  })
}
```

Step 6: Final Application

The completed Shiny app allows users to select a department, and the chart updates automatically to show admission results by gender.